Models of random spectra with missing eigenlevels

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Abstract. The power spectrum analysis of stochastic spectra has emerged as a powerful tool for studying both system-specific and universal properties of complex systems. In the context of complex quantum systems, it reveals whether the corresponding classical dynamics is regular or chaotic, or a mixture of both, and encodes a 'degree of chaoticity'. In combination with other long- and short-range spectral fluctuation measures, it provides an effective way to identify system symmetries, determine a degree of incompleteness of experimentally measured spectra, and get the clues about systems' internal structure. Due to experimental limitations, it is virtually impossible to measure complete sequences of eigenlevels. Incompleteness of spectra inevitably influences its statistical properties.

The main objective of this project is to introduce and study a statistical model of missing eigenlevels. Of particular interest is the question of how a failure to measure complete spectral sequences affects two major spectral statistics: the power spectrum and the form factor.

The project is performed under the guidance of Professor Eugene Kanzieper.

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1. Introduction

1.1. Billiards: Classical vs Quantum and Regular vs Chaotic

Quantum chaology [1] studies spectral properties of tiny dynamical systems whose size is so small that their quantum nature becomes essential and thus cannot be disregarded. Ballistic quantum dots of a nano-scale size, which can be fabricated and controlled in laboratories [2], represent a paradigmatic example of such systems.

From the point of view of researchers – mathematicians and physicists – a *classical billiard* is a dynamical system in which a particle (ball) moves along a straight line and gets specularly reflected from a wall without loss of speed. The angle of incidence before the collision is equal to the angle of reflection after collision with a wall (the law of reflection). The nature and properties of a ball trajectory are dictated by the law of reflection and by a specific form of a billiard.

There are two types of billiards [3, 4]: ordered (regular) and chaotic. A shape of a *regular* billiard possesses a high symmetry (think of a circle or a square). Following a ball trajectory in an ordered billiard, one discovers that it looks perfectly regular.

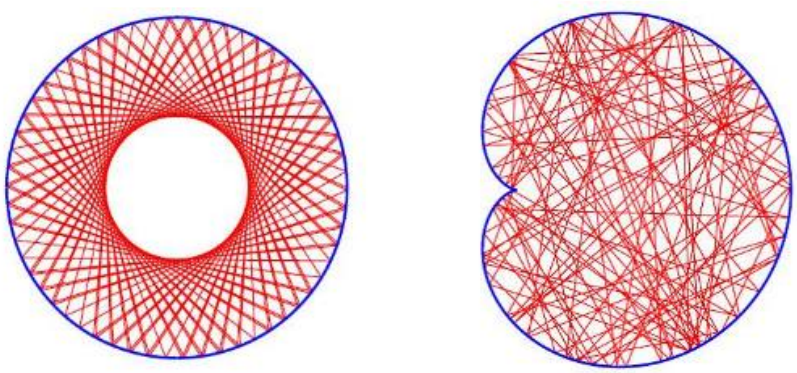

Figure 1. Regular geodesics of a ball in a round (*regular*) billiard and chaotic geodesics of a ball in a cardioid (*chaotic*) billiard. Reproduced from Ref. [5].

On the contrary, a shape of a *chaotic* billiard – exemplified by a cardioid billiard in the figure above – has a lower symmetry, if any. In this case, a ball trajectory appears to be irregular, and a particle tends to uniformly explore the entire billiard. In addition, a distance between two balls with a tiny difference in their initial velocities will diverge very rapidly (exponentially) in time. These two properties are a hallmark of classical chaos.

What will happen if one reduces a billiard size to such an extent that a ball acquires properties of a quantum particle (e.g., starts to behave as an electron described by a wave function)? A similar question: What will happen if one replaces a particle billiard with a wave billiard (e.g., a microwave billiard)? Will it be now possible to determine – through various measurements – whether a quantum (or wave) billiard has an ordered or a chaotic shape? This is one of the major problems in the field of quantum chaos. It is important both theoretically (in quantum mechanics, the Heisenberg uncertainty principle makes the idea of a particle trajectory inappropriate) and practically (modern nanotechnologies make it possible to produce and control quantum billiards).

One of the ways to distinguish between two types of billiards – ordered or chaotic – is to examine statistical properties of energy levels of a quantum particle evolving in a billiards. This can be done by studying either their spectral form factor [4] or a power

spectrum [6], or both. To make a sensible comparison between billiards of various forms and types, one has to eliminate an influence of their system-specific properties encoded in the mean level density $\varrho_N(\lambda)$ of the *measured* energy levels $\{\lambda_1, \ldots, \lambda_N\}$. This is achieved by means of the unfolding procedure [7] represented by the map

$$\varepsilon_\ell = \int_{-\infty}^{\lambda_\ell} d\lambda \varrho_N(\lambda), \quad \ell = 1, \ldots, N. \tag{1.1}$$

It is the unfolded energy levels $\{\varepsilon_1, \ldots, \varepsilon_N\}$ that may obey universal statistical laws as $N \to \infty$.

As far as the power spectrum is concerned, earlier numerical simulations [8] have suggested that the power spectrum of energy levels in regular quantum billiards produce a noise similar to that of a classical particle performing Brownian motion. In contrast, in chaotic billiards fluctuating energy levels are anticipated to produce a white-like noise. A heuristic explanation of these observations was given in Ref. [9].

1.2. Complete vs Incomplete Spectra and Project Objective

A nonperturbative theory of the form factor and the power spectrum for both regular and chaotic billiards was formulated very recently in Refs. [10, 11, 12]. These studies have assumed that *all* eigenlevels belonging to an energy interval of interest can be measured in an experiment. This is certainly an oversimplification since – due to experimental limitations – it is virtually impossible to measure *complete* sequences of eigenlevels; a part of them will inevitably be missed [13].

How will this fact affect our knowledge of a quantum system? The main objective of this project is to formulate a mathematical model of incomplete spectrum and study how a failure to measure complete spectral sequences may affect two spectral statistics: the spectral form factor and the power spectrum of quantum billiards.

The project will focus on analytical calculation of both spectral statistics for spectral sequences with independent level spacings which are known to mimick fluctuations of energy levels in quantum systems with *integrable classical dynamics*. This being said, we shall attempt to formulate a general theory to make it potentially applicable to many more spectral models, e.g., ensembles of random diagonal matrices and/or random matrices of the Wigner-Dyson type [7].

2. Complete Spectrum: Overview

In this Section, we provide basic definitions of spectral sequences with independent level spacings, the spectral form factor (SFF) and the power spectrum (PS), formulate central assumptions regarding the fluctuating properties of random spectra and review major results for both SFF and PS in case of complete spectra.

2.1. Definitions

2.1.1. Spectral Sequences with Independent Level Spacings

Definition 2.1. Let $\{s_1, \cdots, s_N\}$ be independent and identically distributed, positive-definite random variables describing *uncorrelated level spacings* and let $\varepsilon_\ell = \sum_{\ell=1}^{N} s_\ell$ for all $\ell = 1, \cdots, N$. Then the sequence $\{0 \leq \varepsilon_1 < \cdots < \varepsilon_N\}$ is called eigenlevel sequence with i.i.d. spacings.

Remark 2.2. In what follows we shall set the mean level spacing $\langle s_\ell \rangle = 1$ and denote the variance $\mathrm{var}[s_\ell] = \sigma^2$ for all $\ell = 1, \cdots, N$.

2.1.2. Power Spectrum

Definition 2.3. Let $\{\varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ be a sequence of ordered unfolded eigenlevels, $N \in \mathbb{N}$, with the mean level spacing Δ and let $\langle \delta\varepsilon_\ell \delta\varepsilon_m \rangle$ be the covariance matrix of level displacements $\delta\varepsilon_\ell = \varepsilon_\ell - \langle \varepsilon_\ell \rangle$ from their mean $\langle \varepsilon_\ell \rangle = \ell\Delta$. A Fourier transform of the covariance matrix

$$S_N(\omega) = \frac{1}{N\Delta^2} \sum_{\ell=1}^{N} \sum_{m=1}^{N} \langle \delta\varepsilon_\ell \delta\varepsilon_m \rangle \, e^{i\omega(\ell-m)}, \quad \omega \in \mathbb{R} \tag{2.1}$$

is called the power spectrum of a sequence. Here, the angular brackets stand for an average over an ensemble of eigenlevel sequences. ■

Remark 2.4. Since the power spectrum is 2π-periodic, real and even function in ω,

$$S_N(\omega + 2\pi) = S_N(\omega), \quad S_N^*(\omega) = S_N(\omega), \quad S_N(-\omega) = S_N(\omega), \tag{2.2}$$

it is sufficient to consider it in the interval $0 \leq \omega \leq \omega_{\mathrm{Ny}}$, where $\omega_{\mathrm{Ny}} = \pi$ is the Nyquist frequency. ■

Remark 2.5. It readily follows from Definition 2.3 that, by tuning the frequency ω in the power spectrum, one may attend to spectral correlations between either adjacent or distant eigenlevels. Indeed, at 'large' frequencies, $\omega = \mathcal{O}(N^0)$ yet below the Nyquist frequency, the distant eigenlevels barely contribute to the power spectrum; characterized by large values of $|\ell - m|$, they produce strongly oscillating terms in Eq. (2.1) which effectively cancel each other. As the result, $S_N(\omega)$ is mainly shaped by correlations between the nearby levels. At low frequencies, $\omega \ll 1$, these oscillations are by far less pronounced thus making a contribution of distant eigenlevels increasingly important. ■

2.1.3. Spectral Form Factor

Definition 2.6. Let $\{\varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ be a sequence of (not necessarily) ordered unfolded eigenlevels, $N \in \mathbb{N}$, with the mean level spacing Δ. The spectral form factor of such a sequence equals

$$K_N(\tau) = \frac{1}{N} \left(\Big\langle \sum_{\ell=1}^{N} \sum_{m=1}^{N} e^{2i\pi\tau(\varepsilon_\ell - \varepsilon_m)/\Delta} \Big\rangle - \Big\langle \sum_{\ell=1}^{N} e^{2i\pi\tau\varepsilon_\ell/\Delta} \Big\rangle \Big\langle \sum_{m=1}^{N} e^{-2i\pi\tau\varepsilon_m/\Delta} \Big\rangle \right). \tag{2.3}$$

Here, summations run over a set $\{\varepsilon_1, \ldots, \varepsilon_N\}$ of either ordered or unordered unfolded eigenlevels. The angular brackets stand for an average over an ensemble of eigenlevel sequences. ■

2.2. Spectral Stationarity

In what follows we shall assume that random spectra of our interest are either stationary or infinitely-stationary. Both concepts are defined as follows.

Definition 2.7 (Stationarity)**.** Consider an ordered sequence of eigenlevels $\{0 \leq \varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ with $N \in \mathbb{N}$. Let $\{s_1, \ldots, s_N\}$ be the sequence of spacings between consecutive eigenlevels such that $s_\ell = \varepsilon_\ell - \varepsilon_{\ell-1}$ with $\ell = 1, \ldots, N$ and $\varepsilon_0 = 0$. The sequence of level spacings is said to be *stationary* if (i) the average spacing

$$\langle s_\ell \rangle = \Delta \tag{2.4}$$

is independent of $\ell = 1, \ldots, N$ and (ii) the covariance matrix of *spacings* is of the Toeplitz type:

$$\mathrm{cov}(s_\ell, s_m) = I_{|\ell-m|} - \Delta^2 \tag{2.5}$$

for all $\ell, m = 1, \ldots, N$. Here, I_n is a function defined for non-negative integers n. ■

A necessary and sufficient condition for eigenlevel sequences to possess stationary level spacings, proven in Section 3 of Ref. [11], is formulated in Lemma 2.8.

Lemma 2.8. *For $N \in \mathbb{N}$, let $\{0 \leq \varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ be an ordered sequence of unfolded eigenlevels such that $\langle \varepsilon_1 \rangle = \Delta$. Associated sequence of spacings between consecutive eigenlevels is* ***stationary*** *if and only if*

$$\langle (\varepsilon_\ell - \varepsilon_m)^q \rangle = \langle \varepsilon_{\ell-m}^q \rangle \tag{2.6}$$

for $\ell > m$ and both $q = 1$ and $q = 2$.

Remark 2.9. Examples of *finite–N* eigenlevel sequences with stationary spacings include [11, 12] (i) random spectra with independent, identically distributed (i.i.d.) level spacings; (ii) unfolded spectra in ensembles of random diagonal matrices; (iii) spectra of so-called 'tuned' circular ensembles of random matrices. Interestingly, spectral fluctuations in all aforementioned models appear to satisfy the correlation constraint Eq. (2.6) also for $q > 2$. This leads us to define the notion of random spectra with infinitly-stationary level spacings. ■

Definition 2.10 (Infinite stationarity)**.** An ordered sequence of eigenlevels $\{0 \leq \varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ with $N \in \mathbb{N}$ is said to possess an ***infinitly-stationary*** sequence of level spacings if the relation

$$\langle (\varepsilon_\ell - \varepsilon_m)^q \rangle = \langle \varepsilon_{\ell-m}^q \rangle \tag{2.7}$$

holds for $\ell > m$ and *all* $q \in \mathbb{N}$. ■

Corollary 2.11. *Spectral sequence with independent level spacings, Definition 2.1, possesses infinite stationarity of level spacings.*

Proof. As the ℓ-th ordered eigenlevel is a sum of ℓ i.i.d. random variables, $\varepsilon_\ell = \sum_{j=1}^{\ell} s_j$, both the l.h.s. and r.h.s. in Eq. (2.7) represent the q-th moment of a sum of $(\ell - m)$ i.i.d. random variables, $q \in \mathbb{N}$. Infinite stationarity follows immediately. □

2.3. What is Known?

2.3.1. Power Spectrum

For complete sequences of eigenlevels with *stationary* level spacings, the power spectrum can be represented as a differential operator acting on a generating function of eigenlevel variances, see Ref. [11]:

Theorem 2.12. *Let $N \in \mathbb{N}$ and $0 \leq \omega \leq \pi$. The power spectrum for an eigenlevel sequence $\{0 \leq \varepsilon_1 \leq \ldots \leq \varepsilon_N\}$ with stationary spacings equals*

$$S_N(\omega) = \frac{1}{N\Delta^2} \mathrm{Re} \left(z \frac{\partial}{\partial z} - N - \frac{1 - z^{-N}}{1 - z} \right) \sum_{\ell=1}^{N} \mathrm{var}[\varepsilon_\ell]\, z^\ell, \tag{2.8}$$

where $z = e^{i\omega}$, Δ is the mean level spacing, and

$$\mathrm{var}[\varepsilon_\ell] = \langle \delta \varepsilon_\ell^2 \rangle \tag{2.9}$$

is the variance of ℓ-th ordered eigenlevel.

Yet another useful representation, discovered in Ref. [11] and presented in Theorem 2.13 below, expresses the power spectrum in terms of a generating function of probabilities $E_N(\ell; \epsilon)$ to observe exactly ℓ eigenlevels *below* the energy ε,

$$E_N(\ell; \varepsilon) = \frac{N!}{\ell!(N-\ell)!} \left(\prod_{j=1}^{\ell} \int_0^{\varepsilon} d\epsilon_j \right) \left(\prod_{j=\ell+1}^{N} \int_{\varepsilon}^{\infty} d\epsilon_j \right) P_N(\epsilon_1, \ldots, \epsilon_N). \tag{2.10}$$

Here, $P_N(\epsilon_1, \ldots, \epsilon_N)$ is the joint probability density (JPDF) of N unordered eigenlevels taken from a positive definite spectrum; it is assumed to be symmetric under a permutation of its arguments.

Theorem 2.13. *Let $N \in \mathbb{N}$ and $0 \leq \omega \leq \pi$, and let $\Phi_N(\varepsilon;\zeta)$ be the generating function*

$$\Phi_N(\varepsilon;\zeta) = \sum_{\ell=0}^{N} (1-\zeta)^\ell E_N(\ell;\varepsilon) \tag{2.11}$$

of the probabilities defined in Eq. (2.10). The power spectrum, Definition 2.3, for an eigenlevel sequence with stationary spacings equals

$$S_N(\omega) = \frac{2}{N\Delta^2}\mathrm{Re}\left(z\frac{\partial}{\partial z} - N - \frac{1-z^{-N}}{1-z}\right)\frac{z}{1-z}\int_0^\infty d\epsilon\, \epsilon\left[\Phi_N(\epsilon;1-z) - z^N\right] - \tilde{S}_N(\omega), \tag{2.12}$$

where $z = 1-\zeta = e^{i\omega}$, Δ is the mean level spacing, and

$$\begin{aligned}\tilde{S}_N(\omega) &= \frac{1}{N}\mathrm{Re}\left(z\frac{\partial}{\partial z} - N - \frac{1-z^{-N}}{1-z}\right)\sum_{\ell=1}^{N}\ell^2 z^\ell \\ &= \frac{1}{N}\left|\frac{1-(N+1)z^N + Nz^{N+1}}{(1-z)^2}\right|^2 .\end{aligned} \tag{2.13}$$

Remark 2.14. Notably, representations Eqs. (2.11) and (2.12) suggest that the power spectrum is determined by *spectral correlation functions of all orders.* Contrary to the spacing distribution, which is essentially determined by the gap formation probability [7] $E_N(0;\varepsilon)$, the power spectrum depends on the *entire set* of probabilities $E_N(\ell;\varepsilon)$ with $\ell = 0, 1, \ldots, N$. ■

In the particular case of eigenlevel sequences with independent level spacings, see Definition 2.1, the power spectrum can be obtained immediately out of Theorem 2.12 after noticing that $\mathrm{var}[\varepsilon_\ell] = \sigma^2\ell$, see Ref. [11].

Theorem 2.15. *The power spectrum of eigenlevel sequence with i.i.d. level spacings equals*

$$S_N(\omega) = \frac{2N+1}{4N}\frac{\sigma^2}{\sin^2(\omega/2)}\left(1 - \frac{1}{2N+1}\frac{\sin\left((N+1/2)\omega\right)}{\sin(\omega/2)}\right), \tag{2.14}$$

where $0 \leq \omega \leq \pi$.

Corollary 2.16. *For a set of discrete frequencies $\omega_k = 2\pi k/N$, the power spectrum reduces to*

$$S_N(\omega_k) = \frac{\sigma^2}{2\sin^2(\omega_k/2)}, \quad 0 < \omega_k \leq \pi. \tag{2.15}$$

Remark 2.17. Notice that Eqs. (2.14) and (2.15) for the power spectrum of eigenlevel sequences with uncorrelated level spacings hold *universally.* Indeed, both expressions appear to be independent of a particular choice of the level spacings distribution; the level spacing variance σ^2 is the only model-specific parameter. ■

Remark 2.18. Remarkably, in the case of exponentially distributed level spacings ($\sigma^2 = 1$), the power spectrum prediction Eq. (2.15) was recently observed [12] in quantum irrational rectangular billiards.

2.3.2. Spectral Form Factor

In the particular case of eigenlevel sequences with independent level spacings, the spectral form factor can be calculated from Definition 2.3. The following Theorem holds [11].

Theorem 2.19. *Let*

$$\Psi_s(\tau) = \langle e^{2i\pi\tau\, s_\ell} \rangle = \int_0^\infty ds\, e^{2i\pi\tau s} f_{s_\ell}(s) \tag{2.16}$$

be a characteristic function of ℓ-th level spacing whose probability density function is $f_{s_\ell}(s)$. Then, the spectral form factor equals

$$K_N(\tau) = 1 + \frac{2}{N}\mathrm{Re}\left[\frac{\Psi_s(\tau)}{1-\Psi_s(\tau)}\left(N - \frac{1-\Psi_s^N(\tau)}{1-\Psi_s(\tau)}\right)\right] - \frac{1}{N}\left|\Psi_s(\tau)\frac{1-\Psi_s^N(\tau)}{1-\Psi_s(\tau)}\right|^2. \tag{2.17}$$

Corollary 2.20. *For eigenlevel sequences with exponentially distributed level spacings, the spectral form factor equals*

$$K_N(\tau) = 1 + \frac{1}{4\pi^2\tau^2 N}\left(1 - \frac{1}{(1+4\pi^2\tau^2)^N}\right) - \frac{1}{\pi\tau N}\frac{\sin[N\arctan(2\pi\tau)]}{(1+4\pi^2\tau^2)^{N/2}}, \tag{2.18}$$

where $\tau \geq 0$.

Proof. Substitute

$$\Psi_s(\tau) = \int_0^\infty ds\, e^{-s(1-2i\pi\tau)} = \frac{1}{1-2i\pi\tau} \tag{2.19}$$

into Eq. (2.17). □

Remark 2.21. The above prediction Eq. (2.18) was recently observed [12] in quantum irrational rectangular billiards.

3. Probabilistic Model of Incomplete Spectra

To account for eigenlevel which failed to be measured in an experiment, we shall introduce a probailistic model which assumes that incomplete (thinned) spectrum contains precisely n eigenlevels out of $N \geq n$ eigenlevels of the complete spectrum.

3.1. The Model

Let $\{0 \leq \varepsilon_1 \leq \varepsilon_2 \leq \cdots \leq \varepsilon_N\}$ be a sequence of ordered, unfolded and randomly fluctuating eigenlevels. Out of it, we shall construct a subsequence of n ordered eigenlevels ($1 \leq n \leq N$) by choosing a subset $\mathcal{J} = \{1 \leq j_1 < j_2 < \cdots < j_n \leq N\}$ of n of *random indices* out of the complete set $\{1, 2, \cdots, N\}$. By simple combinatorial argument, the joint probability function of n random indices equals

$$P_{n,N}(j_1 = i_1, j_2 = i_2, \cdots, j_n = i_n) = \frac{1}{C_N^n}\mathbb{1}_{1\leq i_1<i_2<\cdots<i_n\leq N}, \tag{3.1}$$

where $C_N^n = \binom{N}{n}$. All partial probability functions can readily be derived from Eq. (3.1).

Lemma 3.1. *The ℓ-th ordered index is a* $\mathrm{NegHyp}(N,n,\ell)$ *random variable described by the probability function*

$$P_{n,N}(j_\ell = k) = \frac{C_{k-1}^{\ell-1} C_{N-k}^{n-\ell}}{C_N^n} \tag{3.2}$$

defined on

$$\Omega_{n,N}(\ell;k) = \{\ell \le k \le \ell + N - n;\ 1 \le \ell \le n \le N\}\,. \tag{3.3}$$

Proof. By definition,

$$P_{n,N}(j_\ell = k) = \sum_{\substack{i_1,\cdots,i_{\ell-1}\\ i_{\ell+1},\cdots,i_n}} P_{n,N}(j_1 = i_1,\cdots,j_{\ell-1} = i_{\ell-1}; j_\ell = k; j_{\ell+1} = i_{\ell+1},\cdots,j_n = i_n). \tag{3.4}$$

Having in mind that

$$\sum_{i_1,\cdots,i_{\ell-1}=1}^{k-1} \mathbb{1}_{1\le i_1<\cdots<i_{\ell-1}\le k-1} = C_{k-1}^{\ell-1} \tag{3.5}$$

and, likewise,

$$\sum_{i_{\ell+1},\cdots,i_n=k+1}^{N} \mathbb{1}_{k+1\le i_{\ell+1}<\cdots<i_n\le N} = C_{N-k}^{n-\ell}, \tag{3.6}$$

we arrive at the statement of this Lemma. □

Corollary 3.2. *As soon as $j_\ell \sim \mathrm{NegHyp}(N,n,\ell)$ we conclude that*

$$\mathbb{E}[j_\ell] = \ell\frac{N+1}{n+1}, \quad \mathrm{var}[j_\ell] = \frac{(N+1)(N-n)}{(n+1)^2(n+2)}\,\ell(n+1-\ell). \tag{3.7}$$

Lemma 3.3. *The joint probability function of ℓ-th and m-th ordered indices ($\ell < m$) equals*

$$P_{n,N}(j_\ell = k, j_m = k') = \frac{C_{k-1}^{\ell-1} C_{k'-k-1}^{m-\ell-1} C_{N-k'}^{n-m}}{C_N^n} \tag{3.8}$$

and is defined on

$$\Omega_{n,N}(\ell,m;k,k') = \{1 \le \ell \le k \le k' + \ell - m \le N + \ell - n;\ 1 \le \ell < m \le n \le N\}\,. \tag{3.9}$$

Proof. By definition,

$$P_{n,N}(j_\ell = k, j_m = k') = \sum_{\substack{i_1,\cdots,i_{\ell-1}\\ i_{\ell+1},\cdots,i_{m-1}\\ i_{m+1},\cdots,i_n}} P_{n,N}\left(\begin{array}{c} j_1 = i_1,\cdots,j_{\ell-1} = i_{\ell-1}; j_\ell = k;\\ j_{\ell+1} = i_{\ell+1},\cdots,j_{m-1} = i_{m-1}; j_m = k';\\ j_{m+1} = i_{m+1},\cdots,j_n = i_n \end{array}\right). \tag{3.10}$$

Complemented with three counting identities

$$\sum_{i_1,\cdots,i_{\ell-1}=1}^{k-1} \mathbb{1}_{1\le i_1<\cdots<i_{\ell-1}\le k-1} = C_{k-1}^{\ell-1}, \tag{3.11}$$

$$\sum_{i_{\ell+1},\cdots,i_{m-1}=k+1}^{k'-1} \mathbb{1}_{k+1\le i_{\ell+1}<\cdots<i_{m-1}\le k'-1} = C_{k'-k-1}^{m-\ell-1}, \tag{3.12}$$

$$\sum_{i_{m+1},\cdots,i_n=k'+1}^{N} \mathbb{1}_{k'+1\le i_{m+1}<\cdots<i_n\le N} = C_{N-k'}^{n-m}, \tag{3.13}$$

the summation in Eq. (3.10) can be readily done. This completes the proof. □

In full generality, the following holds.

Theorem 3.4. *The joint probability function of q ordered indices is given by the formula:*

$$P_{n,N}(j_{\ell_1}=k_1, j_{\ell_2}=k_2,\cdots,j_{\ell_q}=k_q) = \frac{1}{C_N^n} C_{k_1-1}^{\ell_1-1}\left(\prod_{j=2}^{q} C_{k_j-k_{j-1}-1}^{\ell_j-\ell_{j-1}-1}\right) C_{N-k_q}^{n-\ell_q}. \tag{3.14}$$

Here

$$1\le q\le n, \quad 1\le \ell_1<\ell_2<\cdots<\ell_q\le n, \quad 1\le k_1<k_2<\cdots<k_q\le N. \tag{3.15}$$

In addition, there exists a remarkable sum rule:

Theorem 3.5. *The probability function $P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q)$ satisfies the sum rule*

$$\sum_{1\le\ell_1<\cdots<\ell_q\le n} P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q) = \frac{(N-q)!}{N!}\frac{n!}{(n-q)!}. \tag{3.16}$$

Proof. We start with performing summation in Eq. (3.14) over $1\le\ell_1<\ell_2$. The relevant part reads:

$$\begin{aligned}\sum_{\ell_1=1}^{\ell_2-1} C_{k_1-1}^{\ell_1-1} C_{k_2-k_1-1}^{\ell_2-\ell_1-1} &= \sum_{\ell_1=1}^{\ell_2-1}\binom{k_1-1}{\ell_1-1}\binom{k_2-k_1-1}{\ell_2-\ell_1-1}\\ &= \sum_{j=0}^{\ell_2-2}\binom{k_1-1}{j}\binom{k_2-k_1-1}{\ell_2-2-j} = \binom{k_2-2}{\ell_2-2},\end{aligned} \tag{3.17}$$

where we have used the identity [Prudnikov, vol. 1, § 4.2.5, Eq. (13)]

$$\sum_{j=0}^{n}\binom{a}{j}\binom{b}{n-j} = \binom{a+b}{n}. \tag{3.18}$$

As the result, we conclude that

$$\sum_{1\le\ell_1<\ell_2} P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q) = \frac{1}{C_N^n} C_{k_2-2}^{\ell_2-2}\left(\prod_{j=3}^{q} C_{k_j-k_{j-1}-1}^{\ell_j-\ell_{j-1}-1}\right) C_{N-k_q}^{n-\ell_q}. \tag{3.19}$$

Performing further summation over $2 \le \ell_2 < \ell_3$, we earn

$$\sum_{1\le \ell_1<\ell_2<\ell_3} P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q) = \frac{1}{C_N^n} C_{k_3-3}^{\ell_3-3}\left(\prod_{j=4}^{q} C_{k_j-k_{j-1}-1}^{\ell_j-\ell_{j-1}-1}\right) C_{N-k_q}^{n-\ell_q}. \tag{3.20}$$

A step before the last one brings

$$\sum_{1\le \ell_1<\cdots<\ell_{q-1}<\ell_q} P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q) = \frac{1}{C_N^n} C_{k_q-q}^{\ell_q-q} C_{N-k_q}^{n-\ell_q}. \tag{3.21}$$

Finally, summation over $q \le \ell_q \le n$ yields

$$\begin{aligned}\sum_{1\le \ell_1<\cdots<\ell_{q-1}<\ell_q\le n} P_{n,N}(j_{\ell_1}=k_1,\cdots,j_{\ell_q}=k_q) &= \frac{1}{C_N^n}\sum_{\ell_q=q}^{n} C_{k_q-q}^{\ell_q-q} C_{N-k_q}^{n-\ell_q} \\ &= \frac{C_{N-q}^{n-q}}{C_N^n} = \frac{(N-q)!}{N!}\frac{n!}{(n-q)!}. \end{aligned}\tag{3.22}$$

End of proof. □

3.2. Infinite Stationarity of Level Spacings in Incomplete Spectra

A natural question to ask is this: *Given infinite stationarity of level spacings in the complete spectrum, will it also hold in the incomplete spectrum?* Fortunately, this question can be answered in affirmative.

Theorem 3.6. *Infinite stationarity of level spacings of complete spectrum implies infinite stationarity of level spacings in incomplete spectrum.*

Before we turn to the proof of the Theorem 3.6, the following proposition is required:

Proposition 3.7. *Let j_ℓ and j_m be the ℓ-th and m-th ordered indices ($\ell < m$). Then,*

$$j_m - j_\ell \sim j_{m-\ell} \sim \mathrm{NegHyp}(N,n,m-\ell). \tag{3.23}$$

Proof. Let us calculate the probability function of the random variable $j_m - j_\ell$.

$$\begin{aligned} P(j_m - j_\ell = k) &= \sum_{\Omega_{n,N}(\ell,m;j,j+k)} P_{n,N}(j_\ell = j, j_m = k+j) \\ &= \frac{C_{k-1}^{m-\ell-1}}{C_N^n}\sum_{j=\ell}^{N+m-n-k}\binom{j-1}{\ell-1}\binom{N-k-j}{n-m}. \end{aligned}\tag{3.24}$$

The sum over j can be evaluated with the help of the identity [Prudnikov, vol. 1, § 4.2.5, Eq. (37)]

$$\sum_{s=0}^{M}\binom{s+a}{a}\binom{b-s}{b-M} = \binom{a+b+1}{M}. \tag{3.25}$$

We obtain:

$$\begin{aligned}\sum_{j=\ell}^{N+m-n-k}\binom{j-1}{\ell-1}\binom{N-k-j}{n-m} &= \sum_{s=0}^{N+m-n-k-\ell}\binom{s+\ell-1}{\ell-1}\binom{N-k-\ell-s}{n-m} \\ &= \binom{N-k}{N-k-n+m-\ell} = \binom{N-k}{n-(m-\ell)}. \end{aligned}\tag{3.26}$$

This yields:

$$P(j_m - j_\ell = k) = \frac{C_{k-1}^{(m-\ell)-1} C_{N-k}^{n-(m-\ell)}}{C_N^n} \equiv P_{n,N}(j_{m-\ell} = k) \tag{3.27}$$

defined on $\Omega_{n,N}(m-\ell;k)$. This ends the proof. □

At this point we are ready to prove Theorem 3.6.

Proof. According to Definition 2.10, we need to prove that

$$\langle(\hat{\varepsilon}_m - \hat{\varepsilon}_\ell)^q\rangle = \langle\hat{\varepsilon}_{m-\ell}^q\rangle \tag{3.28}$$

holds for all $q \in \mathbb{N}$ and $\ell < m$. Here, the ℓ-th ordered eigenlevel $\hat{\varepsilon}_\ell$ of incomplete spectrum is the j_ℓ-th ordered eigenlevel ε_{j_ℓ} of the complete spectrum, that is $\hat{\varepsilon}_\ell = \varepsilon_{j_\ell}$. Indeed,

$$\begin{aligned}\langle(\hat{\varepsilon}_m - \hat{\varepsilon}_\ell)^q\rangle &= \mathbb{E}_{\{\varepsilon\}}\left[\mathbb{E}_{\{j_\ell,j_m\}}\left[(\varepsilon_{j_m} - \varepsilon_{j_\ell})^q\right]\right] \\ &= \mathbb{E}_{\{\varepsilon\}}\left[\sum_{\Omega_{n,N}(\ell,m;k,k')} (\varepsilon_{k'} - \varepsilon_k)^q P_{n,N}(j_\ell = k, j_m = k')\right] \\ &= \sum_{\Omega_{n,N}(\ell,m;k,k')} \mathbb{E}_{\{\varepsilon\}}\left[(\varepsilon_{k'} - \varepsilon_k)^q\right] P_{n,N}(j_\ell = k, j_m = k').\end{aligned} \tag{3.29}$$

However, infinite stationarity of the complete spectrum implies that

$$\mathbb{E}_{\{\varepsilon\}}\left[(\varepsilon_{k'} - \varepsilon_k)^q\right] = \mathbb{E}_{\{\varepsilon\}}\left[\varepsilon_{k'-k}^q\right]. \tag{3.30}$$

Therefore,

$$\begin{aligned}\langle(\hat{\varepsilon}_m - \hat{\varepsilon}_\ell)^q\rangle &= \sum_{(k,k')\in\Omega_{n,N}(\ell,m)} \mathbb{E}_{\{\varepsilon\}}\left[\varepsilon_{k'-k}^q\right] P_{n,N}(j_\ell = k, j_m = k') \\ &= \mathbb{E}_{\{\varepsilon\}}\left[\mathbb{E}_{\{j_\ell,j_m\}}\left[\varepsilon_{j_m-j_\ell}^q\right]\right].\end{aligned} \tag{3.31}$$

Owing to Proposition 3.7, the random variables $j_m - j_\ell$ and $j_{m-\ell}$ are identically distributed. Therefore,

$$\mathbb{E}_{\{\varepsilon\}}\left[\mathbb{E}_{\{j_\ell,j_m\}}\left[\varepsilon_{j_m-j_\ell}^q\right]\right] = \mathbb{E}_{\{\varepsilon\}}\left[\mathbb{E}_{\{j_{m-\ell}\}}\left[\varepsilon_{j_{m-\ell}}^q\right]\right] = \langle\hat{\varepsilon}_{m-\ell}^q\rangle. \tag{3.32}$$

This ends the proof. □

3.3. Simple Statistical Measures of Incomplete Spectra

In this part of my work, I shall concentrate on finding useful relations between mean level spacing, mean spectral density and spectral correlation functions in complete and incomplete spectra.

3.3.1. Mean Level Spacing

Due to stationarity of level spacings in incomplete spectrum, proven in Theorem 3.6, the mean level spacing $\Delta' = \langle \hat{\varepsilon}_\ell - \hat{\varepsilon}_{\ell-1} \rangle$ does *not* depend on ℓ. Therefore, we shall first focus on the mean value of the ℓ-th ordered eigenlevel therein:

$$\begin{aligned}\langle \hat{\varepsilon}_\ell \rangle &= \mathbb{E}_{\{\varepsilon\}} \left[\mathbb{E}_{\{j_\ell\}} \left[\varepsilon_{j_\ell} \right] \right] = \mathbb{E}_{\{\varepsilon\}} \left[\sum_{\Omega_{n,N}(\ell;k)} \varepsilon_k P_{n,N}(j_\ell = k) \right] \\ &= \sum_{\Omega_{n,N}(\ell;k)} \mathbb{E}_{\{\varepsilon\}} \left[\varepsilon_k \right] P_{n,N}(j_\ell = k) \end{aligned} \tag{3.33}$$

Stationarity of spacings in the complete spectrum implies, according to Definition 2.7, that

$$\mathbb{E}_{\{\varepsilon\}} \left[\varepsilon_k \right] = k\Delta,$$

where Δ is the mean level spacing in the complete spectrum. Therefore,

$$\langle \hat{\varepsilon}_\ell \rangle = \Delta \sum_{k \in \Omega_{n,N}(\ell)} k P_{n,N}(j_\ell = k) = \Delta \mathbb{E}[j_\ell] = \ell \frac{N+1}{n+1} \Delta. \tag{3.34}$$

see Eq. (3.7). Consequently,

$$\Delta' = \frac{N+1}{n+1} \Delta. \tag{3.35}$$

3.3.2. Mean Spectral Density

Lemma 3.8. *Let*

$$R_N^{(1)}(\varepsilon) = \mathbb{E}_{\{\varepsilon\}} \left[\sum_{k=1}^{N} \delta(\varepsilon - \varepsilon_k) \right] \tag{3.36}$$

be a mean level density of the complete spectrum and let

$$R_{n,N}^{(1)}(\varepsilon) = \mathbb{E}_{\{\varepsilon\}} \left[\mathbb{E}_{\{j_\ell\}} \left[\sum_{\ell=1}^{n} \delta(\varepsilon - \varepsilon_{j_\ell}) \right] \right] \tag{3.37}$$

be a mean level density of the thinned spectrum. The following holds:

$$R_{n,N}^{(1)}(\varepsilon) = \frac{n}{N} R_N^{(1)}(\varepsilon). \tag{3.38}$$

Proof. Let us calculate the mean density of the incomplete spectrum:

$$\begin{aligned} R_{n,N}^{(1)}(\varepsilon) &= \mathbb{E}_{\{\varepsilon\}} \left[\mathbb{E}_{\{j_\ell\}} \left[\sum_{\ell=1}^{n} \delta(\varepsilon - \varepsilon_{j_\ell}) \right] \right] \\ &= \mathbb{E}_{\{\varepsilon\}} \left[\sum_{\ell=1}^{n} \sum_{k=1}^{N} \delta(\varepsilon - \varepsilon_k) P_{n,N}(j_\ell = k) \right]. \end{aligned} \tag{3.39}$$

Even though the inner summation should run over the domain $\Omega_{n,N}(\ell;k)$, it was extended to $(1, N)$ as the probability function $P_{n,N}(j_\ell = k)$ nullifies automatically away from $\Omega_{n,N}(\ell;k)$; then, the two sums can be interchanged:

$$R_{n,N}^{(1)}(\varepsilon) = \mathbb{E}_{\{\varepsilon\}} \left[\sum_{k=1}^{N} \delta(\varepsilon - \varepsilon_k) \left(\sum_{\ell=1}^{n} P_{n,N}(j_\ell = k) \right) \right]. \tag{3.40}$$

According to Theorem 3.5, the inner sum equals n/N bringing

$$R^{(1)}_{n,N}(\varepsilon) = \frac{n}{N} R^{(1)}_N(\varepsilon). \tag{3.41}$$

Notice that stationarity played no role in the derivation. □

3.3.3. Density-Density Correlation Function

Lemma 3.9. *Let*

$$R^{(2)}_N(\varepsilon, \varepsilon') = \mathbb{E}_{\{\varepsilon\}} \left[\sum_{k \neq k' = 1}^{N} \delta(\varepsilon - \varepsilon_k) \delta(\varepsilon' - \varepsilon_{k'}) \right] \tag{3.42}$$

be a two-point correlation function of the complete spectrum and let

$$R^{(2)}_{n,N}(\varepsilon, \varepsilon') = \mathbb{E}_{\{\varepsilon\}} \left[\mathbb{E}_{\{j_\ell, j_m\}} \left[\sum_{\ell \neq m = 1}^{n} \delta(\varepsilon - \varepsilon_{j_\ell}) \delta(\varepsilon' - \varepsilon_{j_m}) \right] \right] \tag{3.43}$$

be a two-point correlation function of the thinned spectrum. The following relation holds:

$$R^{(2)}_{n,N}(\varepsilon, \varepsilon') = \frac{n(n-1)}{N(N-1)} R^{(2)}_N(\varepsilon, \varepsilon'). \tag{3.44}$$

Proof. To prove Lemma, we write down

$$R^{(2)}_{n,N}(\varepsilon, \varepsilon') = F_{n,N}(\varepsilon, \varepsilon') + F_{n,N}(\varepsilon', \varepsilon), \tag{3.45}$$

where

$$\begin{aligned} F_{n,N}(\varepsilon, \varepsilon') &= \mathbb{E}_{\{\varepsilon\}} \left[\mathbb{E}_{\{j_\ell, j_m\}} \left[\sum_{\ell < m = 1}^{n} \delta(\varepsilon - \varepsilon_{j_\ell}) \delta(\varepsilon' - \varepsilon_{j_m}) \right] \right] \\ &= \mathbb{E}_{\{\varepsilon\}} \left[\sum_{\ell < m = 1}^{n} \sum_{1 \leq k < k' \leq N} \delta(\varepsilon - \varepsilon_k) \delta(\varepsilon' - \varepsilon_{k'}) P_{n,N}(j_\ell = k, j_m = k') \right]. \end{aligned} \tag{3.46}$$

Even though the inner summation should run over the domain $\Omega_{n,N}(\ell, m; k, k')$, it was extended to $1 \leq k < k' \leq N$ as the joint probability function nullifies automatically away from $\Omega_{n,N}(\ell, m; k, k')$. Interchanging the two sums, we obtain:

$$F_{n,N}(\varepsilon, \varepsilon') = \mathbb{E}_{\{\varepsilon\}} \left[\sum_{1 \leq k < k' \leq N} \delta(\varepsilon - \varepsilon_k) \delta(\varepsilon' - \varepsilon_{k'}) \left(\sum_{\ell < m} P_{n,N}(j_\ell = k, j_m = k') \right) \right]. \tag{3.47}$$

According to Theorem 3.5, the inner sum equals $n(n-1)/N(N-1)$ so that

$$F_{n,N}(\varepsilon, \varepsilon') = \frac{n(n-1)}{N(N-1)} \mathbb{E}_{\{\varepsilon\}} \left[\sum_{1 \leq k < k' \leq N} \delta(\varepsilon - \varepsilon_k) \delta(\varepsilon' - \varepsilon_{k'}) \right]. \tag{3.48}$$

Combining this with the decomposition Eq. (3.45) and the definition Eq. (3.42) ends the proof. □

4. Spectral Form Factor and Power Spectrum in Incomplete Spectrum

4.1. Spectral Form Factor

The easiest way to evaluate the spectral form factor as defined in Definition 2.6 is through its alternative representation in terms of Fourier transforms of the mean level density and the density-density correlation function:

$$K_N(\tau) = 1 + K_N^{(1)}(\tau) - K_N^{(2)}(\tau), \tag{4.1}$$

where

$$K_N^{(1)}(\tau) = \frac{1}{N}\int_{-\infty}^{\infty} d\varepsilon \int_{-\infty}^{\infty} d\varepsilon' e^{2i\pi\tau(\varepsilon-\varepsilon')/\Delta} R_N^{(2)}(\varepsilon,\varepsilon') \tag{4.2}$$

whilst

$$K_N^{(2)}(\tau) = \frac{1}{N}\left|\int_{-\infty}^{\infty} d\varepsilon e^{2i\pi\tau\varepsilon/\Delta} R_N^{(1)}(\varepsilon)\right|^2. \tag{4.3}$$

This representation follows from Definition 2.6, Lemma 3.8 and Lemma 3.9.

Lemma 4.1. *Let the spectral form factor of the complete spectrum admits the representation as in Eqs. (4.1)–(4.3) above. The form factor of the incomplete spectrum equals*

$$K_{n,N}(\tau) = 1 + \frac{n-1}{N-1} K_N^{(1)}\left(\tau\frac{n+1}{N+1}\right) - \frac{n}{N} K_N^{(2)}\left(\tau\frac{n+1}{N+1}\right). \tag{4.4}$$

Proof. The form factor of incomplete spectrum is defined as

$$K_{n,N}(\tau) = 1 + K_{n,N}^{(1)}(\tau) - K_{n,N}^{(2)}(\tau), \tag{4.5}$$

where

$$K_{n,N}^{(1)}(\tau) = \frac{1}{n}\int_{-\infty}^{\infty} d\varepsilon \int_{-\infty}^{\infty} d\varepsilon' e^{2i\pi\tau(\varepsilon-\varepsilon')/\Delta'} R_{n,N}^{(2)}(\varepsilon,\varepsilon') \tag{4.6}$$

whilst

$$K_{n,N}^{(2)}(\tau) = \frac{1}{n}\left|\int_{-\infty}^{\infty} d\varepsilon e^{2i\pi\tau\varepsilon/\Delta'} R_{n,N}^{(1)}(\varepsilon)\right|^2. \tag{4.7}$$

Making use of Eq. (3.35) for the renormalized mean level spacing Δ', and Lemma 3.8 and Lemma 3.9, we observe the relations

$$K_{n,N}^{(1)}(\tau) = \frac{n-1}{N-1} K_N^{(1)}\left(\tau\frac{n+1}{N+1}\right) \tag{4.8}$$

and

$$K_{n,N}^{(2)}(\tau) = \frac{n}{N} K_N^{(2)}\left(\tau\frac{n+1}{N+1}\right). \tag{4.9}$$

This ends the proof. □

Corollary 4.2. *Let the spectral form factor $K_N(\tau)$ of the complete spectrum possesses a well defined limit for $\tau = \mathcal{O}(N^0)$ as $N \to \infty$,*

$$\lim_{N\to\infty} K_N(\tau) = K_\infty(\tau) = 1 + K_\infty^{(1)}(\tau) - K_\infty^{(2)}(\tau), \tag{4.10}$$

and let the length $n(N)$ of the incomplete spectrum grows as $N \to \infty$ in such a way that the ratio

$$p = \lim_{N\to\infty} \frac{n(N)}{N} \tag{4.11}$$

is kept fixed $0 < p \leq 1$. *Then the spectral form factor of incomplete sequence is well defined as* $N \to \infty$ *and equals*

$$K_{p,\infty}(\tau) = 1 - p + pK_\infty(p\tau). \tag{4.12}$$

Discussion.

(1) Corollary 4.2 suggests that incompleteness of measured spectra does *generically* affects the spectral form factor by setting *longer time scales*. For one, if the form factor of the complete spectrum had a specific feature popping up at times τ_*, the same feature will appear in the spectral form factor of the measured (incomplete) spectrum at times

$$\tau_*^{(p)} = \frac{\tau_*}{p} \geq \tau_*. \tag{4.13}$$

(2) Let us stress that this statement however does *not* apply to a large class of quantum billiards with *integrable* classical dynamics introduced and discussed in the Introduction. Indeed, such billiards are described by the model of random spectral sequences with independent, exponentially distributed spacings, for which the spectral form factor is given by Corollary 2.20. Indeed, as $N \to \infty$ and $\tau = \mathcal{O}(N^0)$, Eq. (2.18) yields $K_\infty(\tau) = 1$. By virtue of Corollary 4.2, we then have

$$K_{p,\infty}(\tau) = K_\infty(\tau) = 1. \tag{4.14}$$

Hence, *incomplete measurements of spectra in quantum billiards whose classical dynamics is regular will not affect the spectral form factor.*

(3) This is *not* the case for quantum billiards with chaotic classical dynamics. There, in absence of the time-reversal symmetry, the spectral form factor of complete spectrum is known to follow the universal law [7]

$$K_\infty(\tau) = \begin{cases} \tau, & 0 \leq \tau < 1; \\ 1, & \tau \geq 1. \end{cases} \tag{4.15}$$

Consequently, Corollary 4.2 yields the form factor of the incomplete spectrum in the form

$$K_{p,\infty}(\tau) = \begin{cases} 1 - p + p^2\tau, & 0 \leq \tau < \dfrac{1}{p}; \\ 1, & \tau \geq \dfrac{1}{p}. \end{cases} \tag{4.16}$$

Hence, *incomplete measurements of spectra in quantum billiards whose classical dynamics is chaotic will affect the spectral form factor, in particular, by setting longer time scales.*

4.2. Power Spectrum

While a general relation between the power spectra of complete and incomplete spectrum can also be established [14], its analysis is quit cumbersome. For this reason, we shall directly consider the power spectrum for an incomplete sequence of eigenlevels produced by random sampling of a complete eigenlevel sequence with i.i.d. spacings.

For that we invoke the First Master formula , but first we calculate the variance of eigenlevels in thinned spectra.

$$var[\epsilon_{j_\ell}] = \langle \epsilon_{j_\ell}^2 \rangle - \langle \epsilon_{j_\ell} \rangle^2 = \langle \epsilon_{j_\ell}^2 \rangle - (\ell \Delta'^2)$$

So we will calculate the missing data:

$$\langle \epsilon_{j_\ell}^2 \rangle = \sum_{m=\ell}^{N-(M-\ell)} \langle \epsilon_m^2 \rangle P(j_\ell = m)$$

Observing the relation: $var[\epsilon_m] = var[\sum_i^m s_i] = m\sigma^2$ and $var[\epsilon_m] = \langle \epsilon_m^2 \rangle - \langle \epsilon_m \rangle^2 = \langle \epsilon_m^2 \rangle - (m\Delta)^2$. Letting $\Delta = 1$ we get $\langle \epsilon_m^2 \rangle = m^2 + m\sigma^2$, so:

$$\sum_{m=\ell}^{N-(M-\ell)} (m^2 + m\sigma^2) P(j_\ell = m) = \langle j_\ell^2 \rangle + \sigma^2 \langle j_\ell \rangle =$$

$$= \ell \frac{(N+1)(N-M)}{(M+1)(M+2)} (1 - \frac{\ell}{M+1}) + \ell^2 (\frac{N+1}{M+1})^2 + \sigma^2 \ell \frac{N+1}{M+1}$$

By using $j_\ell \sim NegHyp(\ell, N, M)$. Hence we get:

$$var[\epsilon_{j_\ell}] = \langle \epsilon_{j_\ell}^2 \rangle - \langle \epsilon_{j_\ell} \rangle^2 = \ell \frac{(N+1)(N-M)}{(M+1)(M+2)} (1 - \frac{\ell}{M+1}) + \ell^2 (\frac{N+1}{M+1})^2 + \sigma^2 \ell \frac{N+1}{M+1} - \ell^2 (\frac{N+1}{M+1})^2$$

$$= \ell \frac{(N+1)(N-M)}{(M+1)(M+2)} (1 - \frac{\ell}{M+1}) + \sigma^2 \ell \frac{N+1}{M+1}$$

Now by **The First Master** theorem we have:

$$S_{N,M}(\omega) = \frac{1}{M} \mathrm{Re}(z \frac{\partial}{\partial z} - N - \frac{1 - z^{-N}}{1 - z}) \sum_{\ell=1}^{N} \frac{var[\epsilon_\ell]}{\Delta'^2} z^\ell$$

$$\frac{var[\epsilon_\ell]}{\Delta'^2} = \frac{M+1}{N+1} (\sigma^2 + \frac{N-M}{M+2}) \ell - \frac{1}{n+2} \frac{N-M}{N+1} \ell^2$$

To put it all together we get the following equation:

$$S_{N,M}(\omega) = \frac{1}{M} \mathrm{Re}(z \frac{\partial}{\partial z} - N - \frac{1 - z^{-N}}{1 - z}) \sum_{\ell=1}^{N} z^\ell \{ \ell A_{M,N}(\sigma^2) - \ell^2 B_{N,M} \}$$

$$A_{M,N}(\sigma^2) = (\sigma^2 + \frac{N-M}{M+2}) \frac{M+1}{N+1}$$

$$B_{N,M} = \frac{1}{M+2} \frac{N-M}{N+1}$$

The following relations (taken from the paper: Annals of Physics 413, 168065) let us conclude with an elegant form:

$$\frac{1}{M}\mathrm{Re}(z\frac{\partial}{\partial z} - N - \frac{1-z^{-M}}{1-z})\sum_{\ell=1}^{M}\ell z^{\ell} = S_M^{(1)}(\omega) = \frac{2M+1}{4M}\frac{\sigma^2}{\sin^2(\omega/2)}(1 - \frac{1}{2M+1}\frac{\sin((M+\frac{1}{2})\omega)}{\sin(\frac{\omega}{2})}$$

$$\frac{1}{M}\mathrm{Re}(z\frac{\partial}{\partial z} - N - \frac{1-z^{-M}}{1-z})\sum_{\ell=1}^{M}\ell^2 z^{\ell} = -(n+1)S_M^{(2)}(\omega)$$

$$S_M^{(2)}(\omega) = -\frac{M}{M+1}\frac{1}{4\sin^2(\omega/2)}(1 - \frac{2\sin(\frac{\omega M}{2})}{M\sin(\omega/2)}\cos(\frac{\omega}{2}(M+1)) + \frac{1}{M^2}\frac{\sin^2(\omega M/2)}{\sin^2(\omega/2)}$$

So all in all we derive the following form:

$$S_{N,M}(\omega) = (\sigma^2 + \frac{N-M}{M+2})\frac{M+1}{N+1}S_M^{(1)}(\omega) + \frac{N-M}{(M+2)(N+1)}S_M^{(2)}(\omega)$$

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